

SMRITI Party Stock Visibility (PSV)

Final Technical & Business Blueprint

Version 1.0.2 Final Freeze

Executive Summary

SMRITI Party Stock Visibility (PSV) is an Inventory Intelligence module within SMRITI Retail OS designed to solve one of the most common challenges faced by apparel, footwear, FMCG, cosmetics, and distribution businesses.

The challenge arises when stock has already been legally sold and invoiced to distributors, dealers, franchisees, or third-party locations, but physically remains at those locations awaiting further sale.

From an accounting and taxation perspective:

- Revenue is recognized
- Tax obligations are recorded
- Stock ownership is transferred
- Inventory is removed from company books

However, operational visibility is lost.

Management often cannot answer critical questions:

- How much stock is still physically lying with distributors?
- How much has actually been sold to end customers?
- Which locations are performing well?
- Which locations have slow-moving inventory?
- Which locations require replenishment?
- Which locations require audit attention?

PSV addresses these questions while keeping SMRITI Core Inventory and Accounting completely untouched.

Design Philosophy

PSV is intentionally designed as an:

Inventory Intelligence Layer

Not an Inventory Accounting Module.

PSV never modifies:

- Core Stock Ledger
- Core Inventory Balances
- Core Accounting Entries
- Tax Records
- Inventory Valuation

SMRITI Core Inventory remains the official system of record.

PSV provides:

- Visibility
- Reconciliation
- Distributor Intelligence
- Sell-Through Analytics
- Operational Monitoring

Architectural Principle

SMRITI Core Inventory
+
PSV Shadow Inventory
=
Complete Distribution Visibility

The shadow inventory layer tracks inventory after legal sale without affecting the core transaction engines.

Business Problem

Example:

Tattly Threads invoices:

1,000 KORA Shirts

to:

ABC Fashion Distributor

After invoicing:

- Ownership transferred
- Revenue booked
- Taxes recorded

But physically:

Mumbai Outlet	600
Pune Outlet	250
Sold	150

Without PSV:

Management loses visibility.

With PSV:

Every location remains visible until final sell-through.

Core Module Structure

Master Documents

SMRITI Party Stock Account

Represents distributor locations.

Stores:

- Customer
- Location Name
- Region
- Zone
- Area Manager
- Contact Details
- Operational Status

Example:

ABC Fashion - Mumbai
ABC Fashion - Pune
ABC Fashion - Surat

SMRITI PSV Settings

System configuration document.

Stores:

- Upload frequency
 - Variance thresholds
 - Health score weights
 - Monitoring settings
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Transaction Documents

SMRITI Party Sales Upload

Purpose:

Weekly distributor sales imports.

Capabilities:

- Excel import
- Period locking
- Duplicate prevention
- Overselling protection
- Variant-level validation

Status Flow:

Draft
↓
Validated
↓
Imported

SMRITI Party Physical Snapshot

Purpose:

Physical stock audits.

Capabilities:

- Physical counting
- Variance calculation

- Approval workflow
- Reconciliation support

Status Flow:

```

Draft
↓
Pending Approval
↓
Approved

```

System Documents

SMRITI Party Stock Ledger Entry

The heart of PSV.

Characteristics:

- Hidden
- Immutable
- System generated
- Audit friendly

Rules:

```

No Edit
No Delete
No Rename

```

Supported Transactions:

```

Opening
Dispatch
Sales
Adjustment
Return

```

SMRITI PSV Activity Log

Tracks:

- Upload activity

- Audit approvals
- Reconciliations
- Health alerts
- Operational events

Severity Levels:

Info
Warning
High
Critical

SMRITI PSV Exception Record

Tracks unresolved issues.

Examples:

- Negative balances
- Reconciliation mismatches
- Operational exceptions

Lifecycle:

Open
↓
Under Review
↓
Resolved

Variant-Based Inventory Model

PSV operates exclusively at Item Variant level.

Example:

Template:

KORA-1383

Variants:

KORA-1383-BLACK-38
KORA-1383-BLACK-40
KORA-1383-BLACK-42
KORA-1383-BLUE-38

Rule:

`item_code` always refers to the Item Variant.

Valid:

KORA-1383-BLACK-38

Invalid:

KORA-1383

All balances, audits, reconciliations, and analytics are performed at Variant level.

Sales Invoice Integration

Sales Invoices include:

`custom_party_stock_account`

When an invoice is submitted:

Sales Invoice
↓
PSV Service
↓
Dispatch Ledger Entry
↓
Distributor Stock Updated

This process is fully automatic.

Ledger Design

PSV uses a simple accounting-style movement model.

Formula:

Balance = SUM(qty)

Example:

Opening	+500
Dispatch	+100
Sales	-40
Adjustment	-10
Balance	550

Benefits:

- Simple
- Fast
- Auditable
- Scalable

Idempotency & Concurrency Protection

Protection Layer 1:

SHA-256 Hash

Protection Layer 2:

Application Validation

Protection Layer 3:

Database Unique Constraint

This prevents:

- Duplicate imports
- Double processing
- Hook duplication
- Concurrent write corruption

Invoice Cancellation Strategy

Traditional Approach:

Block Cancellation

PSV Approach:

Allow Cancellation
↓
Create Exception Record
↓
Pending Reconciliation

Benefits:

- No operational deadlocks
- Full audit trail
- Controlled reconciliation

Opening Balance Strategy

Many distributors already hold stock before PSV deployment.

PSV supports:

Opening Balance Import

Process:

Import
↓
Validation
↓
Opening Ledger Entries
↓
Migration Baseline Created

Historical invoices are not replayed.

Opening balances become the official starting point.

Excel Import Framework

Supported aliases:

Item

Item
SKU
Article
Style
Product

Quantity

Qty
Quantity
Sold Qty
Sales Qty

Date

Date
Sale Date
Sales Date

This accommodates real-world distributor file variations.

Health Monitoring Engine

A scheduled daily monitoring process evaluates system health.

Checks:

Critical

Negative Balances

High

Pending Reconciliations

Warning

Late Uploads
Never Audited Locations

Info

Alert Resolutions

Alert Suppression Engine

Alert Key Format:

Location|AlertType|Item

Examples:

Mumbai|NEGATIVE_BALANCE|KORA-1383-BLACK-38

Pune|LATE_UPLOAD|

Duplicate alerts are suppressed and updated instead of recreated.

Analytics & Reporting

Party Stock Balance

Current distributor inventory.

Reconciliation Report

Expected vs Physical.

Stock Ageing Report

0-30 Days
31-60 Days

61-90 Days
90+ Days

Sell-Through Report

Formula:

$\text{Sales} \div \text{Dispatch} \times 100$

Example:

Dispatch = 1000
Sales = 700

Sell Through = 70%

Executive Dashboard

Executive KPIs

- Total PSV Stock
- Total PSV Stock Value
- Open Exceptions
- Critical Alerts

Distribution KPIs

- Fast Moving Locations
- Slow Moving Locations
- Highest Sell-Through
- Lowest Sell-Through

Risk KPIs

- Negative Balances
 - Pending Reconciliations
 - Late Uploads
 - Audit Overdue
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Testing Coverage

Validated Areas:

- ✓ Opening Balance
- ✓ Sales Invoice Integration
- ✓ Cancellation Exceptions
- ✓ Upload Validation
- ✓ Physical Audit Processing
- ✓ Query Optimization
- ✓ Concurrency Protection
- ✓ Migration Integrity
- ✓ Health Monitoring

Target Result:

9 / 9 Tests Passing

Deployment Strategy

Phase 1

Foundation

- DocTypes
- Permissions
- Custom Fields

Phase 2

Core Engine

- Ledger Engine
- Balance Engine
- Invoice Hooks

Phase 3

Operations

- Upload Module
- Audit Module
- Exception Engine

Phase 4

Intelligence

- Reports
- Dashboard
- Health Monitoring

Phase 5

Pilot Deployment

- Opening Balances
 - Real Distributor Uploads
 - First Reconciliation Cycle
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Future Roadmap

Version 1.1+

- Internal Party Transfers
 - Balance Cache Engine
 - Forecasting
 - Replenishment Suggestions
 - Distributor Scorecards
 - AI Demand Prediction
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Final Status

Module: SMRITI Party Stock Visibility (PSV)

Version: 1.0.2 Final Freeze

Architecture: Approved

Technical Specification: Approved

Implementation Readiness: Approved

Pilot Readiness: Approved

Next Milestone:

First Live Distributor Upload and First Reconciliation Cycle